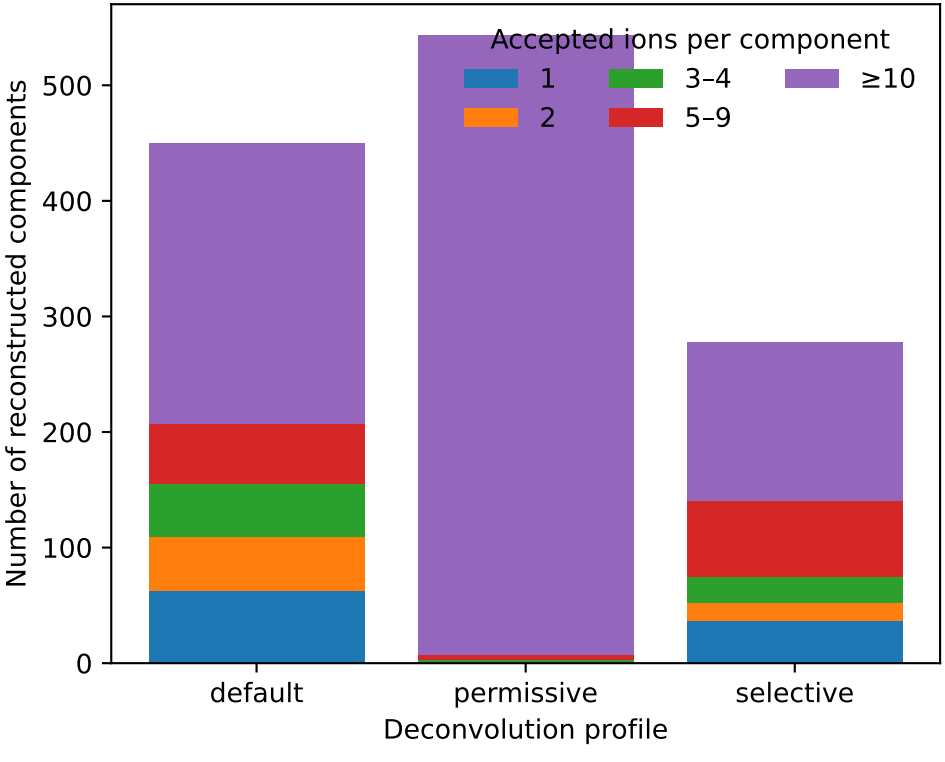

Full-length validation mzML: 7,680 MS1 spectra. Sparse components are retained; automatic NIST submission is separately QC-filtered.